

Hongchi Xia

hongchix@illinois.edu

Siebel School of Computing and Data Science

University of Illinois Urbana-Champaign

Champaign, Illinois

xiahongchi.github.io

Google Scholar

github.com/xiahongchi

Research Interests

Spatial Intelligence for Embodied Physical AI, especially 3D Generation, Reconstruction, and Simulation with Interactive Digital Twins for Embodied AI.

Education

University of Illinois Urbana-Champaign

Ph.D. in Computer Science; Advisors: Prof. Shenlong Wang and Prof. Wei-Chiu Ma

Aug. 2024–present

Champaign, IL

Shanghai Jiao Tong University

B.Eng. in Computer Science and Technology; GPA: 4.07/4.3; Rank: 1/106

Sep. 2020–Jun. 2024

Shanghai, China

Experience

NVIDIA Cosmos Lab

Research Intern

Contributed to Cosmos 3, an omnimodal world model for physical AI.

May 2026–present

Santa Clara, CA

NVIDIA Deep Imagination Research (Now Cosmos Lab)

Research Intern

Developed SAGE for 3D scene generation and contributed to OpenPI Comet.

May 2025–Feb. 2026

Santa Clara, CA

Publications

* denotes equal contribution. My name is bolded.

- [1] **Hongchi Xia**, Tianhang Cheng, Wei-Chiu Ma, Shenlong Wang. “FIRE3D: Feed-forward Interactive 3D Scene Reconstruction Within A Minute.” *Arxiv Preprint*, 2026.
- [2] **Hongchi Xia**, Xuan Li, Zhaoshuo Li, Qianli Ma, Jiashu Xu, Ming-Yu Liu, Yin Cui, Tsung-Yi Lin, Wei-Chiu Ma, Shenlong Wang, Shuran Song, and Fangyin Wei. “SAGE: Scalable Agentic 3D Scene Generation for Embodied AI.” *CVPR*, 2026.
- [3] **Hongchi Xia**, Chih-Hao Lin, Hao-Yu Hsu, Quentin Leboutet, Katelyn Gao, Michael Paulitsch, Benjamin Ummenhofer, and Shenlong Wang. “HoloScene: Simulation-Ready Interactive 3D Worlds from a Single Video.” *NeurIPS*, 2025.
- [4] **Hongchi Xia**, Entong Su, Marius Memmel, Arhan Jain, Raymond Yu, Numfor Mbiziwo-Tiapo, Ali Farhadi, Abhishek Gupta, Shenlong Wang, and Wei-Chiu Ma. “DRAWER: Digital Reconstruction and Articulation With Environment Realism.” *CVPR*, 2025.
- [5] Hao-Yu Hsu, Chih-Hao Lin, Albert J. Zhai, **Hongchi Xia**, and Shenlong Wang. “AutoVFX: Physically Realistic Video Editing from Natural Language Instructions.” *3DV*, 2025.
- [6] **Hongchi Xia**, Zhi-Hao Lin, Wei-Chiu Ma, and Shenlong Wang. “Video2Game: Real-time, Interactive, Realistic and Browser-Compatible Environment from a Single Video.” *CVPR*, 2024.
- [7] **Hongchi Xia***, Yang Fu*, Sifei Liu, and Xiaolong Wang. “RGBD Objects in the Wild: Scaling Real-World 3D Object Learning from RGB-D Video.” *CVPR*, 2024.
- [8] Bo Pang*, **Hongchi Xia***, and Cewu Lu. “Unsupervised 3D Point Cloud Representation Learning by Triangle-Constrained Contrast for Autonomous Driving.” *CVPR*, 2023.

Selected Projects and Open Research Releases

- **Cosmos 3: Omnimodal World Models for Physical AI** *NVIDIA Cosmos Lab, 2026*
Research contributor to an open family of omnimodal world models for jointly processing and generating language, images, video, audio, and action for physical AI.
- **OpenPI Comet: Competition Solution for the 2025 BEHAVIOR Challenge** *NVIDIA, 2025*
Project contributor to a foundation-model adaptation and training system for long-horizon mobile manipulation in the BEHAVIOR Challenge.

Research Experience

Computer Vision Group, University of Illinois Urbana-Champaign Jan. 2023–present
Ph.D. Student; Advisors: Prof. Shenlong Wang and Prof. Wei-Chiu Ma *Champaign, IL*

- Developed Fire3D, a feed-forward system that discovers objects and estimates their 6-DoF poses, then reconstructs complete textured assets in under a minute from an unsegmented image or casual video. [paper][code][project]
- Developed HoloScene, a framework for reconstructing complete, physically stable, simulation-ready interactive worlds from a single video. [paper][code][project]
- Developed DRAWER, an automatic framework for perceiving, reconstructing, articulating, and re-simulating interactive objects in realistic 3D scenes. [paper][code][project]
- Developed Video2Game, which converts casually captured videos into real-time, realistic, browser-compatible game environments and robot simulators. [paper][code][project]

NVIDIA Cosmos Lab May 2026–present
Research Intern; Mentor: Qianli Ma; Manager: Ming-Yu Liu *Santa Clara, CA*

- Contributed to Cosmos 3, an open family of omnimodal world models that jointly processes and generates language, images, video, audio, and action for physical AI. [paper][code][project]

NVIDIA Deep Imagination Research (Now Cosmos Lab) May 2025–Feb. 2026
Research Intern; Mentor: Fangyin Wei; Manager: Ming-Yu Liu *Santa Clara, CA*

- Developed SAGE, a scalable agentic framework that generates simulation-ready 3D scenes from user specifications and supports downstream embodied action generation; released the SAGE-10K dataset. [paper][code][project]
- Contributed to OpenPI Comet, a foundation-model adaptation and training solution for long-horizon mobile manipulation in the 2025 BEHAVIOR Challenge. [paper][code]

Artificial Intelligence Group, University of California San Diego Jun. 2023–Feb. 2024
Visiting Researcher; Advisor: Prof. Xiaolong Wang *San Diego, CA*

- Created the WildRGB-D dataset, comprising approximately 8,500 real-world objects and nearly 20,000 RGB-D videos across 46 categories. [paper][code][project]

Machine Vision and Intelligence Group, Shanghai Jiao Tong University Aug. 2022–Jan. 2023
Research Assistant; Advisor: Prof. Cewu Lu *Shanghai, China*

- Developed Triangle-Constrained Contrast for unsupervised 3D point-cloud representation learning in autonomous-driving scenes. [paper]